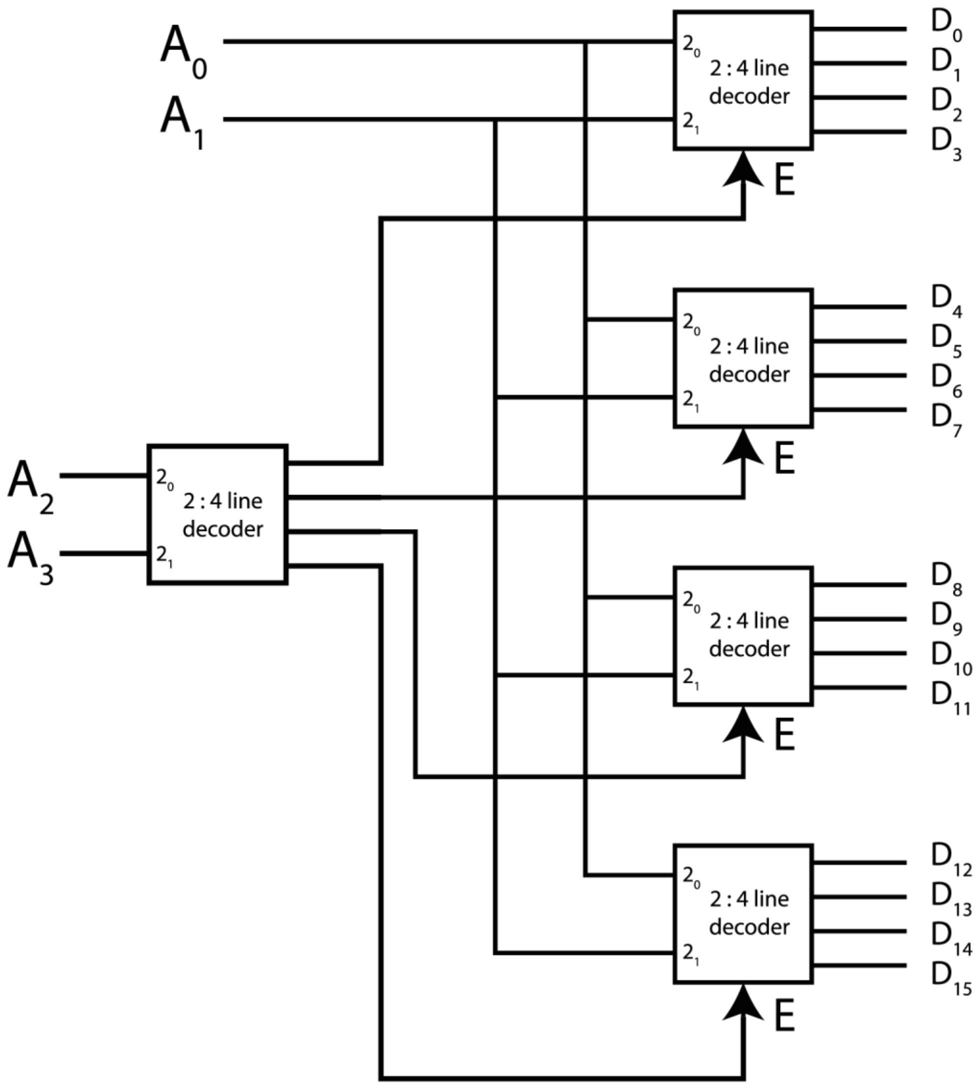
1.

a) Construct a 4-to-16 decoder by joining 2-to-4 decoders with enable input.



Or,

a) Design a multiplexer using the following Boolean Equation.

$$F(A,B,C) = \sum(1,3,4,5,6,7)$$

Steps:

n no. of variables (3)

n-1 no of Section lines (2)

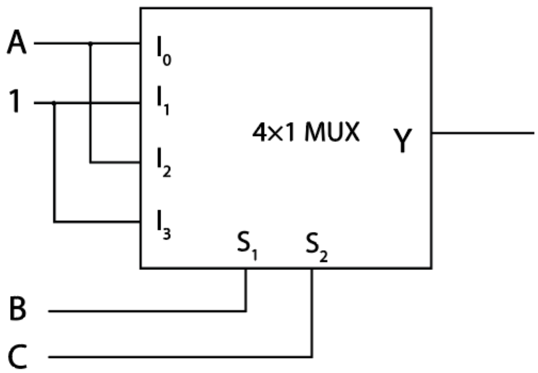
2^n-1: 1 MUX (4:1)

Truth table:

Minterm	A	B	C	F
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

Implement table:

	I_0	I_1	I_2	I_3
A’	0	1	2	3
A	4	5	6	7
	A	1	A	1



b) What is the carry propagation problem in parallel adders? What could be the solution of carry propagation problem of parallel adder?

The carry propagation problem in a parallel adder occurs because each stage of the adder has to wait for the carry bit to be generated by the previous stage, causing a delay.

One solution to this problem is to use a carry-lookahead adder, which reduces the delay by computing carry bits in advance based on the inputs.

B_1	A_1	C_0	S	C_1
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$C_1 = B_1 A_1 C_0 + B_1 A_1 \bar{C}_0 + B_1 \bar{A}_1 C_0 + \bar{B}_1 A_1 C_0$

$= B_1 A_1(C_0 + \bar{C_0}) + C_0(B_1 \bar{A_1} + \bar{B_1} A_1)$

$C_1 = B_1 A_1 + (B_1 \oplus A_1)C_0$

$C_i = \underbrace{B_i A_i}_{\text{Carry generator (Gi)}} + \underbrace{(B_i \oplus A_i)}_{\text{Carry propagator(Pi)}} C_{i-1}$

$C_i = G_i + P_i C_{i-1}$

For $i = 0$;

$C_1 = G_0 + P_0 C_0$

For $i = 1$;

$C_2 = G_1 + P_1 C_1$

$= G_1 + P_1 (G_0 + P_0 C_0)$

$= G_1 + P_1 G_0 + P_1P_0C_0$

For $i = 2$;

$C_3 = G_2 + P_2 C_2$

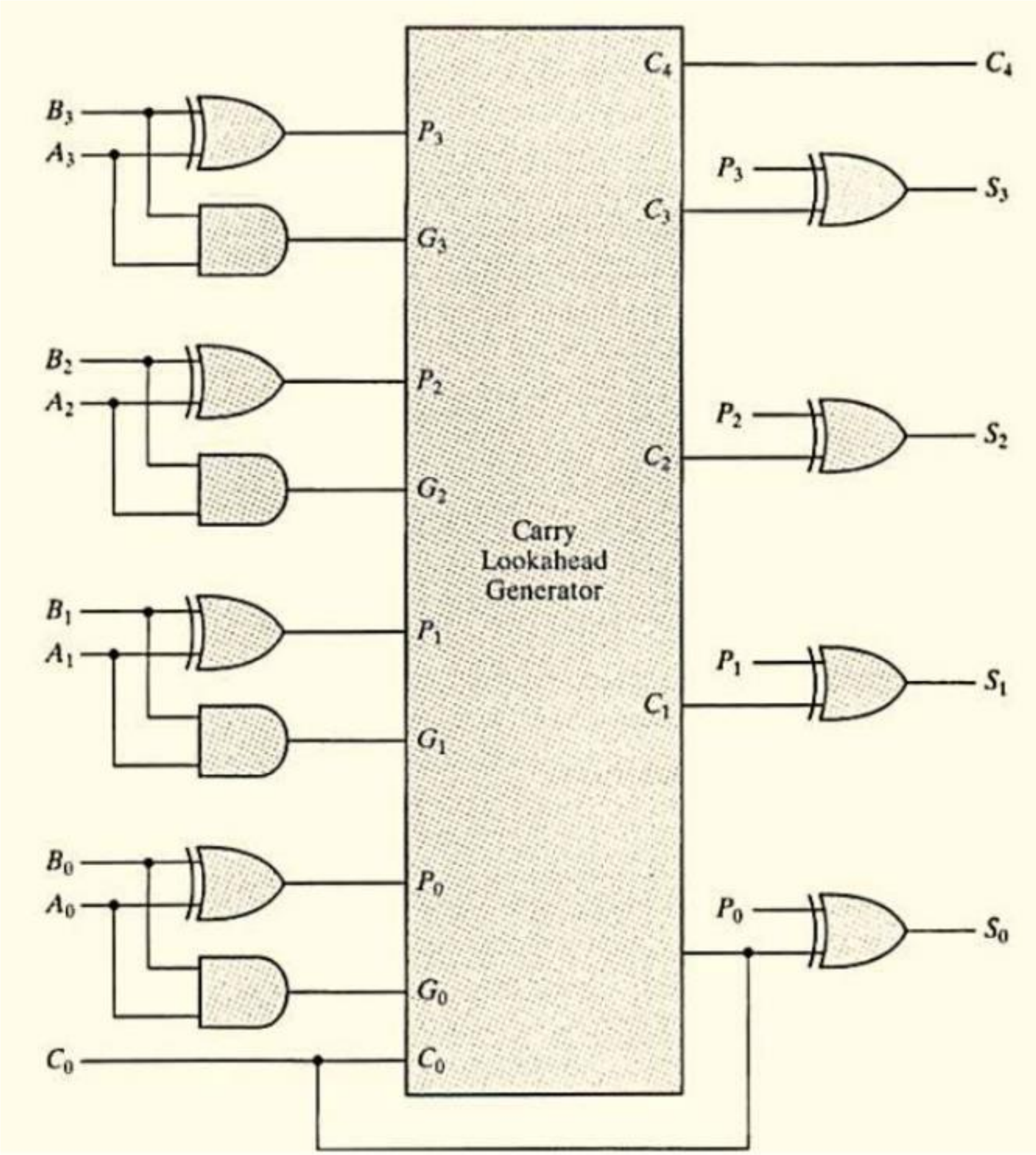
$= G_2 + P_2(G_1 + P_1 G_0 + P_1P_0C_0)$
$= G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2P_1P_0C_0$

For $i = 3$;

$C_4 = G_3 + P_3 C_3$

$= G_3 + P_3(G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2P_1P_0C_0)$

$= G_3 + P_3G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2P_1P_0C_0$



2.

a) Explain the working principle of a JK flip-flop and how it differs from an SR flip-flop. Provide a detailed description of the characteristic table and the excitation table for the JK flip-flop. What is the basic building block of sequential logic circuits?

JK Flip-Flop:

1. **Working Principle:**
   - It has inputs: J, K, Clock (CLK), and Q (output).
   - It toggles when both J and K are HIGH.
2. **Truth Table:**

J	K	Q(n+1)
0	0	Q(n)
0	1	0
1	0	1
1	1	Q(n)’

Difference with SR Flip-Flop:

- JK Flip-Flop resolves the invalid state of SR Flip-Flop when both inputs are HIGH.

JK Flip flop:

Characteristics Table for the JK Flip-Flop:

Q(t)	J	K	Q(t+1)	Comment
0	0	0	0	No change
0	0	1	0	Clear Q
0	1	0	1	Set Q
0	1	1	1	Toggle
1	0	0	1	No change
1	0	1	0	Clear Q
1	1	0	1	Set Q
1	1	1	0	Toggle

Excitation Table for the JK Flip-Flop:

Q(t)	**Q(t+1)**	**J**	**K**
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

The basic building block of sequential logic circuits is the flip-flop**,** as it stores one bit of data and provides memory functionality.

b) Write down the differences between latch and flip flop. Write down the steps associated for SR to JK Flip Flop conversion with proper circuit diagram.

Feature	Latch	Flip-Flop
Triggering Mechanism	Level-triggered	Edge-triggered
Timing	Asynchronous	Synchronous
Control Signal	Enable	Clock
Responsiveness	Changes with enable active	Changes only on clock edge
Common Uses	Simple memory elements	Registers, counters, and more synchronized designs

SR to JK Flip Flop conversion:

Step:1- SR is avaible and JK is required flip flop

Step:02- Characteristic table JK FF

Q_n	J	K	$Q_{(n+1)}$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Step:02- Excitation Table SR FF

Q_t	$Q_{(t+1)}$	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Step:03-

For S

Q \ JK	J'K' 00	J'K 01	JK 11	JK' 10
Q' 0	0	0	1	1
Q 1	X	0	0	X

For R

Q \ JK	J'K' 00	J'K 01	JK 11	10
Q' 0	X	X	0	0
Q 1	0	1	1	0

S = JQ'

R = KQ

Step:04-